

Customer Lifetime Value (CLV) Prediction Using Pareto/NBD Model from Scratch

Abstract

This report presents a Pareto/NBD implementation developed from first principles using NumPy and SciPy's L-BFGS-B optimization algorithm. The model estimates customer purchasing behavior and predicts future transaction frequency and Customer Lifetime Value (CLV) without relying on specialized marketing analytics libraries.

Introduction

Customer Lifetime Value (CLV) is a key metric that estimates the total revenue a business can expect from a customer throughout their relationship. This project implements the Pareto/NBD model from scratch to model customer purchase frequency and dropout behavior.

Theory

The Pareto/NBD model assumes:

Customer transactions follow a Poisson process.

Purchase rates vary across customers according to a Gamma distribution.

Customer lifetimes follow an Exponential distribution.

Dropout rates are Gamma distributed, capturing heterogeneity among customers.

Objectives

Implement the Pareto/NBD model from scratch.

Estimate model parameters using Maximum Likelihood Estimation (MLE).

Predict expected future customer transactions.

Estimate Customer Lifetime Value (CLV).

Methodology

Generate customer transaction summaries (frequency, recency, observation period).

Implement the Pareto/NBD log-likelihood function using NumPy.

Estimate model parameters using SciPy's L-BFGS-B optimizer.

Compute probability of customer being active.

Predict future transactions and calculate projected CLV.

Dataset

Synthetic customer transaction data is generated to validate parameter estimation and prediction accuracy. The implementation can be directly applied to real-world transactional datasets after converting them into customer summary statistics.

Likelihood

The Pareto/NBD parameters are estimated by maximizing the log-likelihood function through Maximum Likelihood Estimation (MLE). Numerical optimization minimizes the negative log-likelihood over all customers.

Complexity Analysis

Vectorized NumPy computations enable efficient $O(N)$ likelihood evaluation for each optimization iteration while minimizing computational overhead by avoiding explicit Python loops.

Results

The implementation successfully estimates purchase-rate and dropout parameters, predicts future customer transactions, computes the probability that customers remain active, and projects Customer Lifetime Value.

Conclusion

The project demonstrates an accurate and computationally efficient implementation of the Pareto/NBD model from scratch. It serves as a practical educational reference and provides a strong foundation for customer analytics and CLV prediction on real-world datasets.

References

Schmittlein, Morrison & Colombo (1987), Counting Your Customers: Who Are They and What Will They Do Next?

Fader, Hardie & Lee (2005), Marketing Science.

SciPy Documentation.

NumPy Documentation.